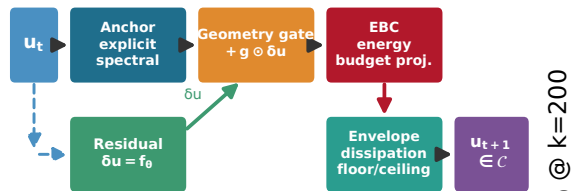


(a) One fused step (v8): Anchor $\rightarrow$ Gate $\rightarrow$ EBC $\rightarrow$ Envelope**Guaranteed per step:**

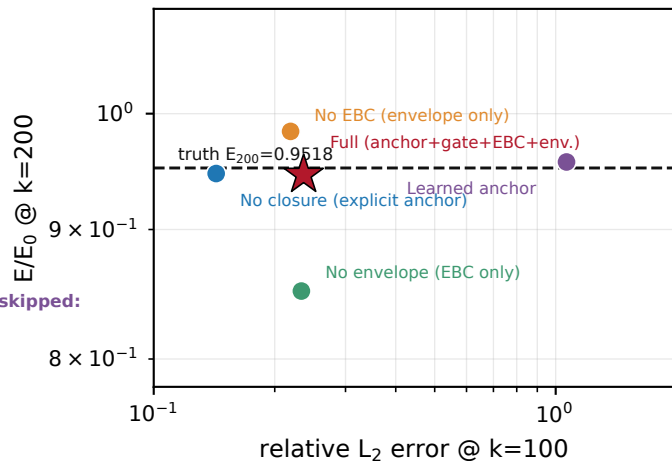
$$\|u_{\text{fused}} - u_{\text{anc}}\| \leq R_{\text{eff}}(u_t)$$

$$E_{\text{in}} - c \epsilon(u_t) \Delta t \leq E(u_{t+1}) \leq E_{\text{in}}$$

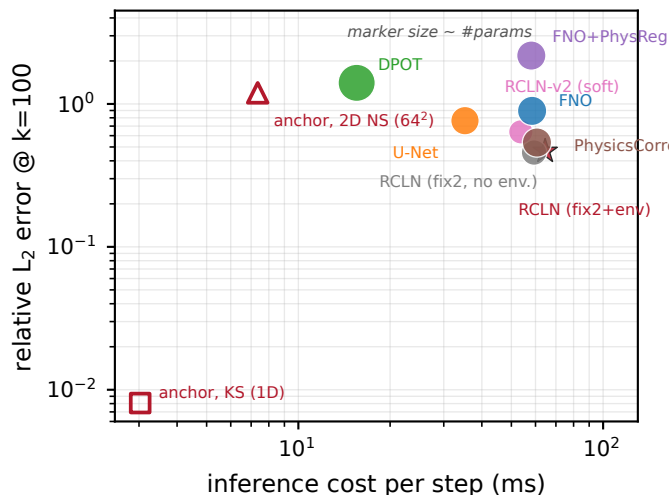
$$\nabla \cdot u_{t+1} = 0 \quad (\text{spectral projection, exact})$$

**constraints cannot be skipped:
every step re-enters C**

(b) Component ablation (3D TGV, v8)



(c) Accuracy-cost Pareto (earlier learned-anchor lineage)

(d) Vorticity PDF @ $k=200$: false vs. true stability